

# LPPool2d#

<https://pytorch.org/docs/stable/generated/torch.nn.LPPool2d.html>

## Description:

Applies a 2D power-average pooling over an input signal composed of several input planes. On each window, the function computed is: At  $p = \infty$ , one gets Max Pooling At  $p = 1$ , one gets Sum Pooling (which is proportional to average pooling) The parameters `kernel_size`, `stride` can either be:

## Function Signature

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```
class torch.nn.LPPool2d(norm_type, kernel_size, stride=None,
                        ceil_mode=False)[source]#
```

Parameters

Shape:

```
forward(input)[source]#
```

Return type

### Returns:

Tensor

## Code Examples

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```
>>> # power-2 pool of square window of size=3, stride=2
>>> m = nn.LPPool2d(2, 3, stride=2)
>>> # pool of non-square window of power 1.2
>>> m = nn.LPPool2d(1.2, (3, 2), stride=(2, 1))
>>> input = torch.randn(20, 16, 50, 32)
>>> output = m(input)
```

## Notes & Warnings

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**NOTE** If the sum to the power of  $p$  is zero, the gradient of this function is not defined. This implementation will set the gradient to zero in this case.

## Detailed Documentation

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### Documentation

Applies a 2D power-average pooling over an input signal composed of several input planes.

On each window, the function computed is:

At  $p = \infty$ , one gets Max Pooling

At  $p = 1$ , one gets Sum Pooling (which is proportional to average pooling)

The parameters `kernel_size`, `stride` can either be:

a single int – in which case the same value is used for the height and width dimension

a tuple of two ints – in which case, the first int is used for the height dimension, and the second int for the width dimension

If the sum to the power of  $p$  is zero, the gradient of this function is not defined. This implementation will set the gradient to zero in this case.

`kernel_size` (Union[int, tuple[int, int]]) – the size of the window

`stride` (Union[int, tuple[int, int]]) – the stride of the window. Default value is `kernel_size`

`ceil_mode` (bool) – when True, will use ceil instead of floor to compute the output shape

Input: (N,C,Hin,Win)(N, C, H<sub>{in}</sub>, W<sub>{in}</sub>)(N,C,Hin,Win) or (C,Hin,Win)(C, H<sub>{in}</sub>, W<sub>{in}</sub>)(C,Hin,Win).

Output: (N,C,Hout,Wout)(N, C, H<sub>{out}</sub>, W<sub>{out}</sub>)(N,C,Hout,Wout) or (C,Hout,Wout)(C, H<sub>{out}</sub>, W<sub>{out}</sub>)(C,Hout,Wout), where

Runs the forward pass.